

DSA1101

Introduction to Data Science

Week 10 Tutorial 8

TODAY'S AGENDA

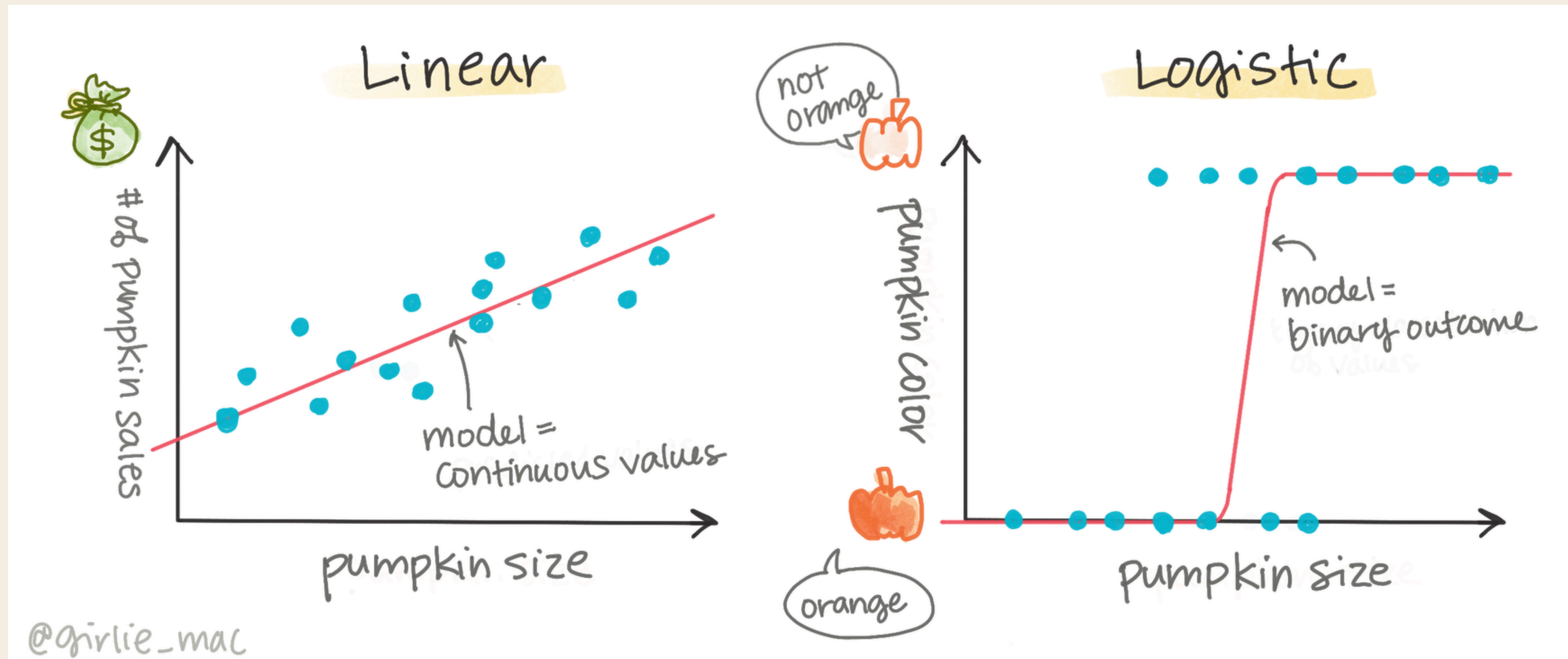
1. Quick Revision
2. On-site Questions Attempt
3. On-site Questions Discussion
4. Off-site Questions Discussion
5. Assignment 2 Tips



QUICK REVISION

LOGISTIC REGRESSION

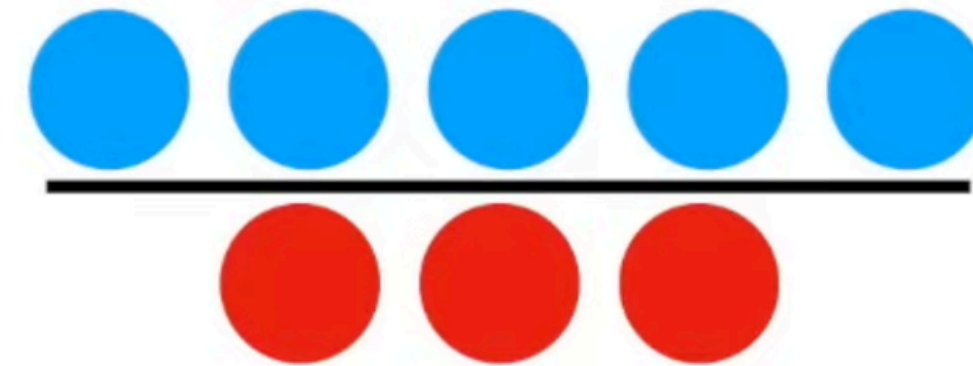
Logistic regression is a linear model for classification that models the log-odds (logit) as a linear function of the features.



ODDS VS PROBABILITY

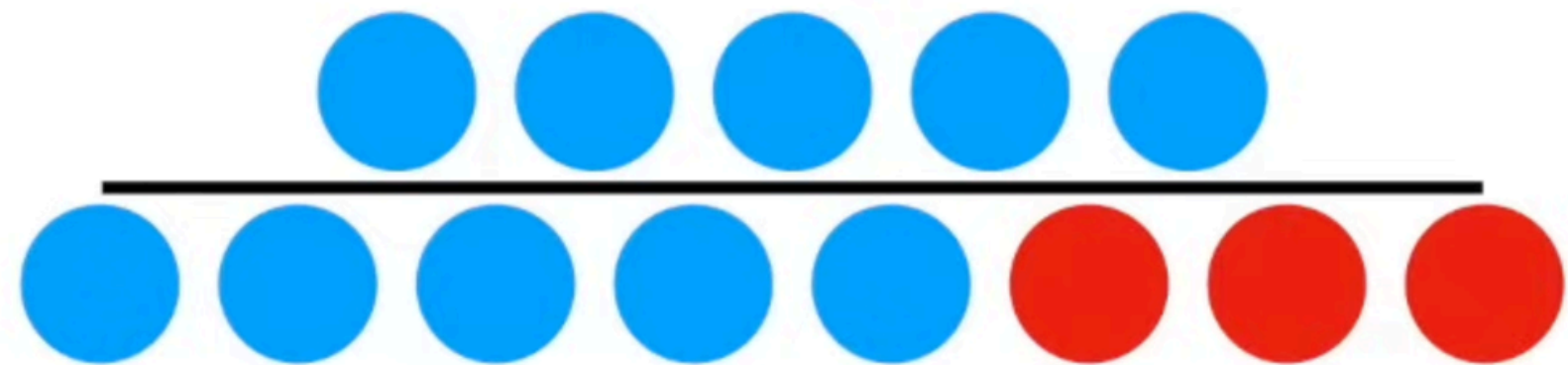
The odds are the ratio of
something happening (i.e. my
team **winning**)...

...to something not happening
(i.e. my team **not winning**).



Probability is the ratio of something
happening (i.e. my team **winning**)...

...to *everything* that could happen
(i.e. my team **winning** and **losing**).



If you do simple math with this example you'll also see

$$\text{Odds} = P(\text{winning}) / P(\text{losing})$$

LOGISTIC REGRESSION? ISN'T IT CLASSIFICATION THOUGH

Modelling Log-odds (Logit) as a linear function of features:

$$\log \left(\underbrace{\frac{P(Y = \text{Class 1} | X)}{1 - P(Y = \text{Class 1} | X)}}_{\text{Odds of Class 1}} \right) = \log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_m X_m$$

- Note: log-odds ranges $(-\infty, +\infty)$.
 - That's why we can use linear model to model it!
 - We're essentially performing "regression" on the log-odds
 - Hence the name Logistic Regression!

SIGMOID FUNCTION (EXTRA-ISH)

Modelling Log-odds (Logit) as a linear function of features:

$$\log \left(\underbrace{\frac{P(Y = \text{Class 1} | X)}{1 - P(Y = \text{Class 1} | X)}}_{\text{Odds of Class 1}} \right) = \log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_m X_m$$

$$p = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_m X_m}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_m X_m}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_m X_m)}}$$

We're essentially putting the linear equation into a sigmoid function!

Sigmoid Function!

$$\frac{1}{1 + e^{-x}}$$

STEPS OF LOGISTIC REGRESSION

USING SIGMOID (EXTRA-ISH)

Step 1: Applying the regression function

$$\mathbf{z} = \mathbf{w}_0 + \mathbf{w}_1\mathbf{x}_1 + \mathbf{w}_2\mathbf{x}_2 + \cdots + \mathbf{w}_n\mathbf{x}_n$$

Step 2: Passing the output into the sigmoid function to get probability [0,1]

$$g(\mathbf{z}) = \frac{1}{(1 + e^{-\mathbf{z}})}$$

Step 3: Choose a threshold to turn probability into a classifier

- Points with estimated probability $\geq$ threshold belongs to the “positive class” (label = 1), else “negative class” (label = 0)
- Eg. threshold = 0.5 (so all points $\geq$ 0.5 we classify as positive)

Logistic Regression in R - glm()

GLM = Generalized Linear Model

There are many generalized linear model, logistic regression is one of them

Calling model:

```
model <- glm(y ~ x, data = data, family = binomial)
```

family = binomial is a must to include!

If you don't include, it's just linear regression (not logistic)

Logistic Regression – link logit

You probably saw this in the lecture code:

```
M1<- glm( Churned ~., data =churn,  
          family = binomial(link ="logit"))
```

It's okay to include it but it's **OPTIONAL**. As the default is already this, it's okay not to include this too!

Logistic Regression - requirement

glm() in R has one special requirement:

It's response variable HAS to be 0 and 1! (or "0" and "1")

Why?

Cause glm() with family = binomial is trying to model $P(Y = 1)$, so it needs to know which outcome is "success" (1) and which is "failure" (0).

Eg. "Good" "Bad" / "Cancer" "No Cancer" (R doesn't know!)

Logistic Regression - requirement

glm() in R has one special requirement:

It's response variable HAS to be 0 and 1! (or “0” and “1”)

Convert categories of response variables into 0 and 1 before calling glm():

```
ifelse(market$Direction == "Up", 1, 0)
```

```
ifelse(market$Direction == "Down", 0, 1)
```

Our best friend ifelse()!

Logistic Regression – Predicting

Step 1: Predict $P(Y = 1)$ using `predict()`.

```
pred = predict(model, newdata = newdata, type = "response")
```

`type = "response"` just means I want the predicted probability $P(Y = 1)$
there's also `type = "link"` that returns the log-odds prediction (default)

Step 2: Classify points into class using threshold

If probability $P(Y = 1) \geq \text{threshold}$ then class response is 1, else 0

```
class_pred = ifelse(pred >= threshold, 1, 0)
```

Writing Fitted Equation

How do we write the fitted equation for this?

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, famil
    data = churn)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.472062   0.132107  26.282  <2e-16 ***
Age        -0.156635   0.004088 -38.318  <2e-16 ***
Married1     0.066430   0.068299   0.973   0.331
Churned_contacts 0.381909   0.027302  13.988  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\log \left(\frac{p}{1-p} \right) = 3.472 - 0.157 \cdot \text{Age} + 0.066 \cdot \mathbb{I}(\text{Married} = 1) + 0.382 \cdot \text{Churned_contacts}$$

Remember it's the log-odds that follows the linear equation!

Interpretation of numerical variables

Eg. Interpret the coefficient of 'Age' in this model

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, family = binomial, data = churn)

Coefficients:
(Intercept)      3.472062
Age             -0.156635
Married1         0.066430
Churned_contacts  0.381909
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\log\left(\frac{p}{1-p}\right) = 3.472 - 0.157 \cdot (n + 1) + \dots$$
$$= 3.472 - 0.157n - 0.157 + \dots$$

Fixing other variables, a one year increase in Age decreases
the **log-odds** of churning by 0.157

Interpretation of numerical variables

Eg. Interpret the coefficient of 'Age' in this model

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, famil
    data = churn)

Coefficients:

```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.472062	0.132107	26.282	<2e-16 ***
Age	-0.156635	0.004088	-38.318	<2e-16 ***
Married1	0.066430	0.068299	0.973	0.331
Churned_contacts	0.381909	0.027302	13.988	<2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$$\frac{p}{1-p} = e^{3.472-0.157 \cdot (n+1)+\dots} = e^{3.472-0.157n-0.157+\dots}$$

new odds / old odds:

$$\frac{e^{3.472-0.157n-0.157+\dots}}{e^{3.472-0.157n+\dots}} = e^{-0.157} = 0.855$$

Fixing other variables, a one year increase in Age multiplies the **odds** of churning by $e^{(-0.157)} = 0.855$ (aka the odds of churning decrease by about 14.5%)

Interpretation of categorical variables

Eg. Interpret the coefficient of 'Married' in this model

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, famil
    data = churn)

Coefficients:

```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.472062	0.132107	26.282	<2e-16	***
Age	-0.156635	0.004088	-38.318	<2e-16	***
Married1	0.066430	0.068299	0.973	0.331	
Churned_contacts	0.381909	0.027302	13.988	<2e-16	***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Fixing other variables, being married (Married = 1) increases the **log-odds** of churning by 0.066 compared to not being married (Married = 0)

Interpretation of categorical variables

Eg. Interpret the coefficient of 'Married' in this model

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, famil
    data = churn)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.472062   0.132107  26.282  <2e-16 ***
Age        -0.156635   0.004088 -38.318  <2e-16 ***
Married1     0.066430   0.068299   0.973   0.331
Churned_contacts 0.381909   0.027302  13.988  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Fixing other variables, being married multiplies the **odds** of churning by $e^{(0.066)} = 1.068$
(aka the odds of churning increase by about 6.8% compared to not being married)

p-value of a coefficient

The p-value tells you whether that variable actually matters in predicting the outcome.

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, famil
    data = churn)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   3.472062   0.132107  26.282  <2e-16 ***
Age          -0.156635   0.004088 -38.318  <2e-16 ***
Married1      0.066430   0.068299   0.973   0.331
Churned_contacts 0.381909   0.027302  13.988  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- Small p-value (< 0.05) $\rightarrow$ the variable matters, keep it
- Large p-value (≥ 0.05) $\rightarrow$ the variable doesn't matter, it could be removed

p-value of a coefficient

The p-value tells you whether that variable actually matters in predicting the outcome.

```
Call:
glm(formula = Churned ~ Age + Married + Churned_contacts, family = binomial, data = churn)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   3.472062   0.132107  26.282  <2e-16 ***
Age          -0.156635   0.004088 -38.318  <2e-16 ***
Married1      0.066430   0.068299   0.973   0.331
Churned_contacts 0.381909   0.027302  13.988  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Here, **Age and Churned_contacts significantly contribute** to whether a person has churned. But **Married does not!**

EVALUATING LOGISTIC REGRESSION

our fav confusion matrix!

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Sometimes Negative and Positive will switch places. So please read carefully before answering!

Accuracy =

$$(TP + TN) / (TP + TN + FP + FN)$$

$$TPR (Recall) = TP / (TP + FN)$$

$$FPR = FP / (FP + TN)$$

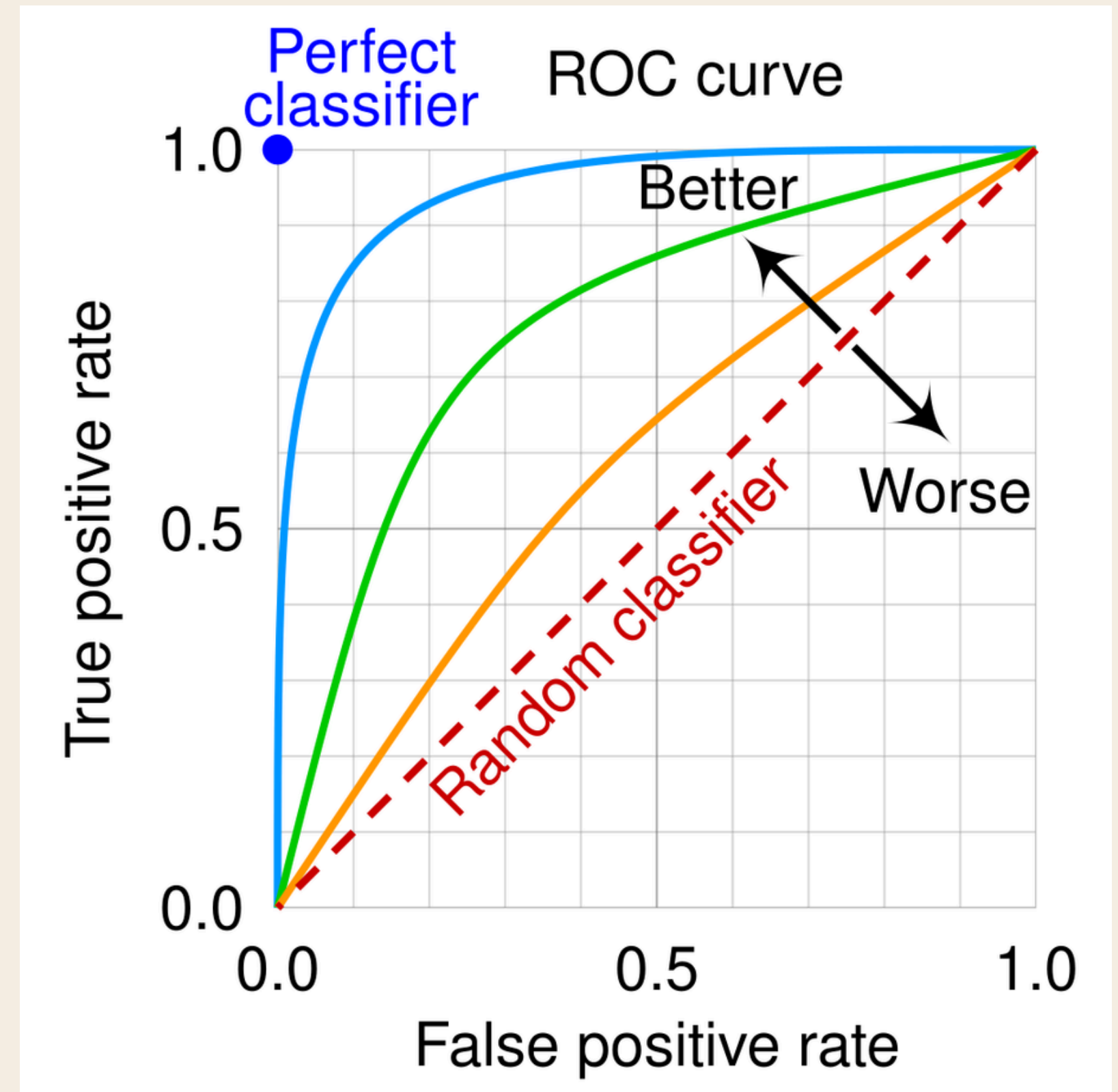
$$FNR = FN / (TP + FN)$$

$$Precision = TP / (TP + FP)$$

NEW EVALUATION TECHNIQUE

ROC Curve

ROC curve plots the TPR and FPR from your confusion matrix at every possible threshold, showing how the tradeoff between them changes as you move the classification cutoff.

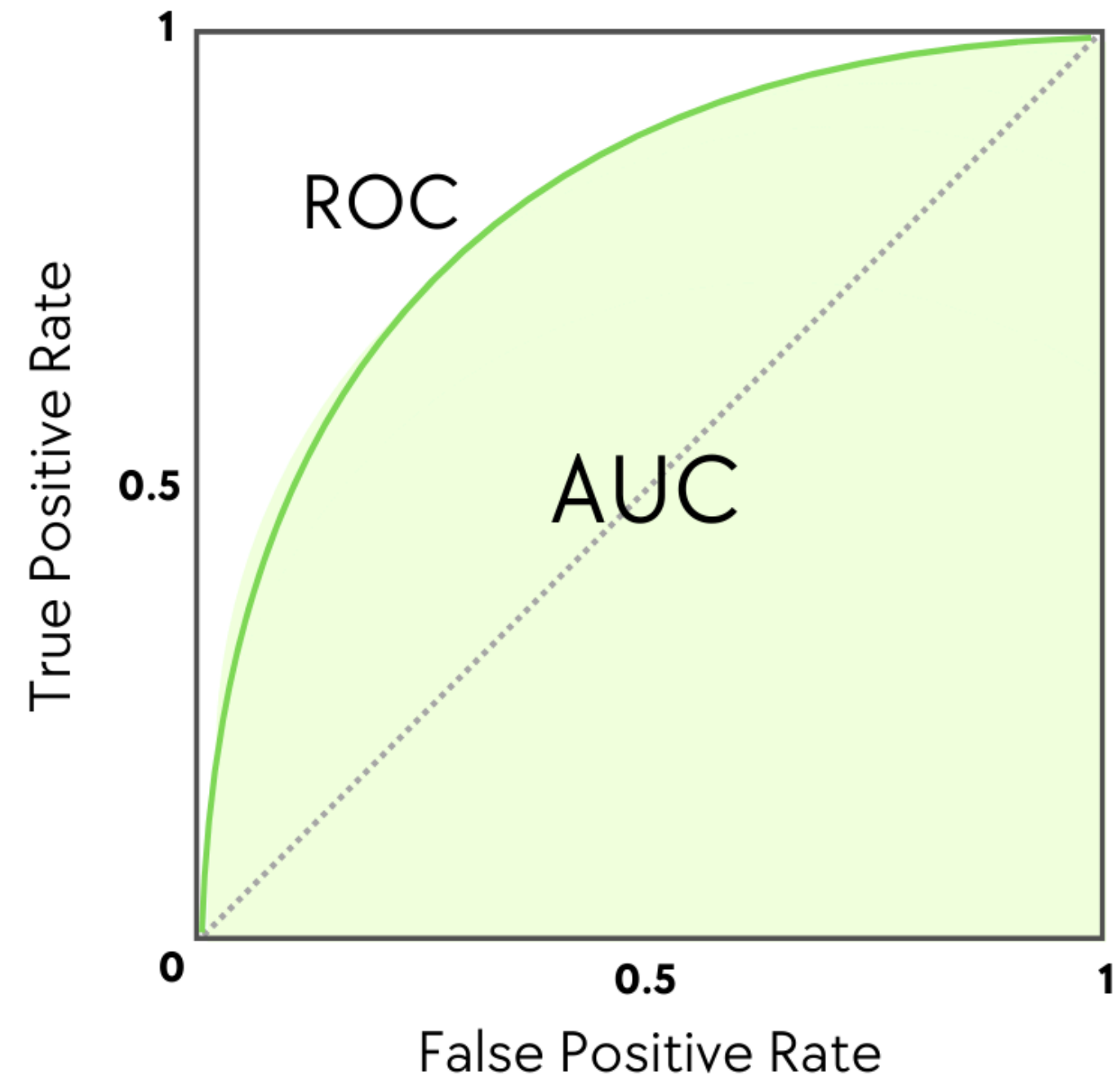


NEW EVALUATION TECHNIQUE

AUC (Area under curve)

Area under the ROC Curve.

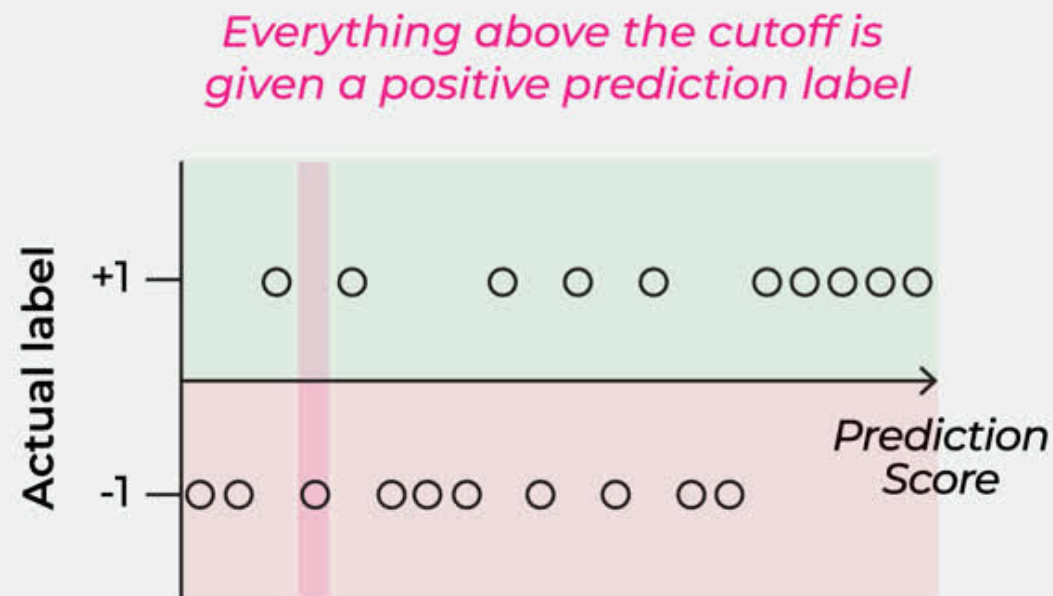
An AUC of 0.5 suggests random guessing, while 1.0 represents a perfect classifier.



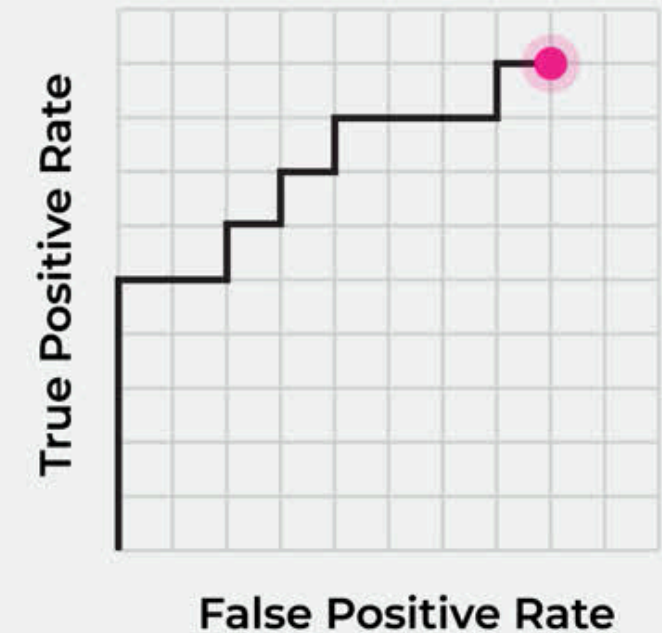
ROC AND AUC

True Positive Rate: 9/10

False Positive Rate: 8/10

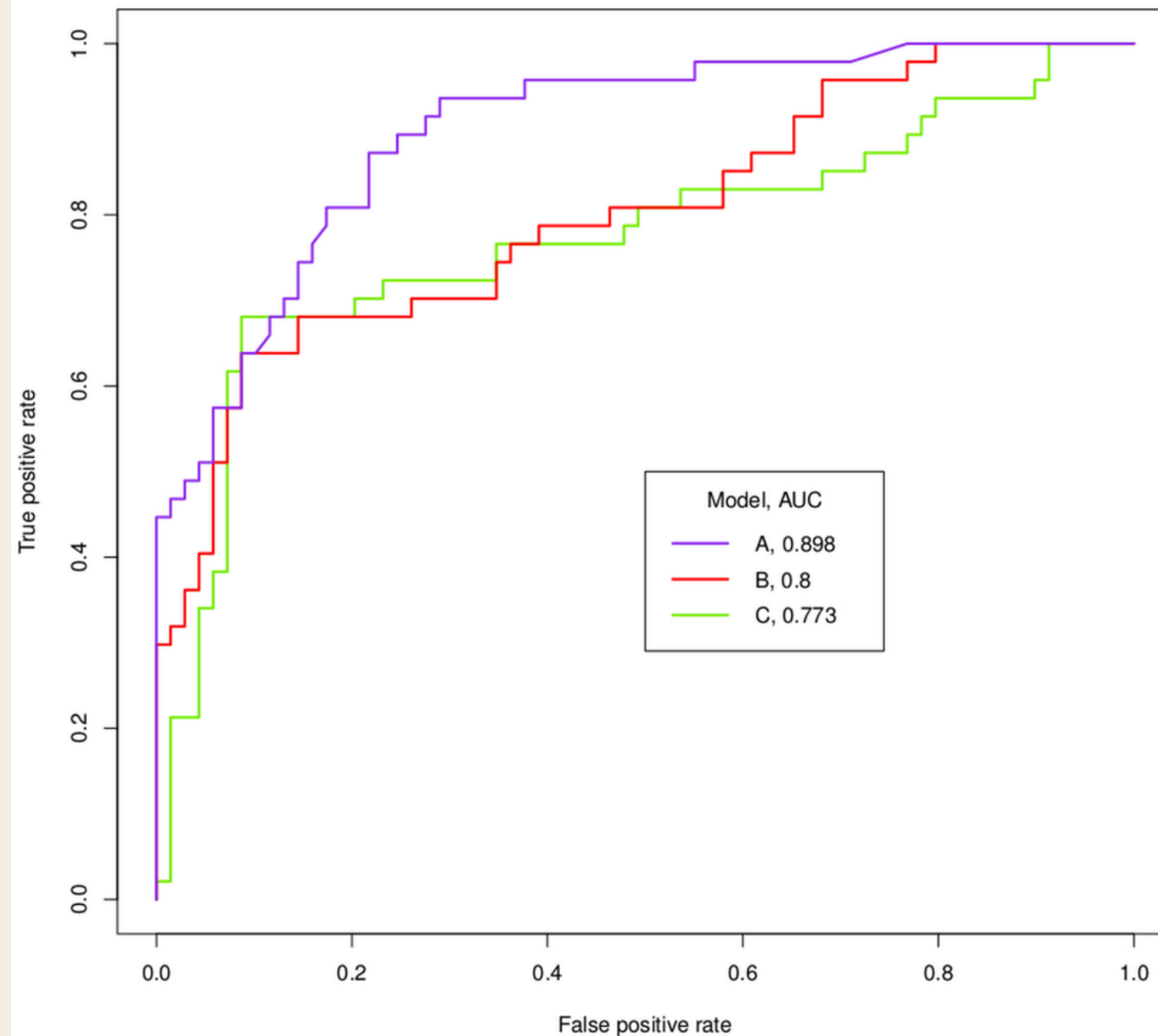


	Prediction label	
	POS (+)	NEG (-)
Actual label		
POS (+)	9	1
NEG (-)	8	2



View GIF here! <https://arize.com/blog/what-is-auc/>

ROC-AUC – VERY GOOD FOR COMPARING MODELS



which is very important for your
upcoming Assignment 2!

ROC-AUC IN R

```
library(ROCR)

# Step 1: Train your model
model <- naiveBayes(Y ~ ., data = traindata)

# Step 2: Get predicted probabilities for positive class
pred_prob <- predict(model, testdata, type = "raw")
score <- pred_prob[, 2] # probability of positive class (2nd column)

# Step 3: Create ROCR prediction object
pred_obj <- prediction(score, testdata$Y) # (predicted scores, true labels)

# Step 4: Compute TPR and FPR
roc <- performance(pred_obj, measure = "tpr", x.measure = "fpr")

# Step 5: Plot ROC curve
plot(roc, col = "red")

# Step 6: Compute AUC
auc <- performance(pred_obj, measure = "auc")
auc@y.values[[1]] # extract AUC value
```

ON-SITE QUESTIONS

ON-SITE QUESTIONS

1. Data set `Smarket.csv` contains data on percentage returns for the S&P 500 stock index over 1250 days, from the beginning of 2001 until the end of 2005. For each date, the data contains the trading volume and the percentage returns for each of the five previous trading days, Lag1 through Lag5.
 - (a) Find the proportion of the days that the direction is up.
 - (b) Fit a logistic regression model (called model M3) that helps to predict the direction of S&P 500 index, based on trading volume and the percentage returns for each of the five previous trading days, Lag1 through Lag5.

Discuss with your group! Time limit: 4 mins

ON-SITE PREP AND INSPECT

```
readLines("Data/Smarket.csv", n = 3)
```

```
market = read.csv("Data/Smarket.csv")
```

```
head(market)
```

```
str(market)
```

ON-SITE PREP AND INSPECT

```
'data.frame': 1250 obs. of 10 variables:
 $ X      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Year   : int  2001 2001 2001 2001 2001 2001 2001 2001 2001 2001 ...
 $ Lag1   : num  0.381 0.959 1.032 -0.623 0.614 ...
 $ Lag2   : num  -0.192 0.381 0.959 1.032 -0.623 ...
 $ Lag3   : num  -2.624 -0.192 0.381 0.959 1.032 ...
 $ Lag4   : num  -1.055 -2.624 -0.192 0.381 0.959 ...
 $ Lag5   : num   5.01 -1.055 -2.624 -0.192 0.381 ...
 $ Volume : num  1.19 1.3 1.41 1.28 1.21 ...
 $ Today  : num  0.959 1.032 -0.623 0.614 0.213 ...
 $ Direction: chr  "Up" "Up" "Down" "Up" ...
```

Remember glm() only accepts 0 and 1 (or “0” and “1”) as outputs!

ON-SITE PREP AND INSPECT

Converting “Up” to 1, and the rest (aka “Down”) to 0

```
market$y = ifelse(market$Direction== "Up", 1, 0)
```

Here we create a new column in market called “y”

Question: Is it okay if we do this?

```
market$Direction = ifelse(market$Direction== "Up", 1, 0)
```

Ans: Yes! It’s totally fine, it’ll just overwrite the current column. Here, we create a new column to have record for both

ON-SITE QUESTION 1

Q: Find the proportion of the days that the direction is up.

```
prop.table(table(market$Direction))
```

```
> table(market$Direction)
```

Down	Up
602	648

```
> prop.table(table(market$Direction))
```

Down	Up
0.4816	0.5184

You can also use the new column “y” that we created!

Ans: 51.84% of the days that direction is Up

ON-SITE QUESTION 2

Q: Fit a logistic regression model (M3) that predicts the direction of S&P 500 index, based on **trading volume** and the percentage returns for each of the five previous trading days, **Lag1 through Lag5**

```
> str(market)
'data.frame':   1250 obs. of  11 variables:
 $ X      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Year   : int  2001 2001 2001 2001 2001 2001 2001 2001 2001 2001 ...
 $ Lag1   : num  0.381 0.959 1.032 -0.623 0.614 ...
 $ Lag2   : num  -0.192 0.381 0.959 1.032 -0.623 ...
 $ Lag3   : num  -2.624 -0.192 0.381 0.959 1.032 ...
 $ Lag4   : num  -1.055 -2.624 -0.192 0.381 0.959 ...
 $ Lag5   : num   5.01 -1.055 -2.624 -0.192 0.381 ...
 $ Volume : num   1.19 1.3 1.41 1.28 1.21 ...
 $ Today  : num   0.959 1.032 -0.623 0.614 0.213 ...
 $ Direction: chr  "Up" "Up" "Down" "Up" ...
 $ y      : num   1 1 0 1 1 1 0 1 1 1 ...
```

ON-SITE QUESTION 2

Q: Fit a logistic regression model (M3) that predicts the direction of S&P 500 index, based on **trading volume** and the percentage returns for each of the five previous trading days, **Lag1 through Lag5**

```
M3 <- glm(y ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,  
          data = market,  
          family = binomial)
```

```
summary(M3)
```

1. formula (y ~ x)
2. data
3. family = binomial

ON-SITE QUESTION 2

Q: Fit a logistic regression model (M3) that predicts the direction of S&P 500 index, based on **trading volume** and the percentage returns for each of the five previous trading days, **Lag1 through Lag5**

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.126000	0.240736	-0.523	0.601
Lag1	-0.073074	0.050167	-1.457	0.145
Lag2	-0.042301	0.050086	-0.845	0.398
Lag3	0.011085	0.049939	0.222	0.824
Lag4	0.009359	0.049974	0.187	0.851
Lag5	0.010313	0.049511	0.208	0.835
Volume	0.135441	0.158360	0.855	0.392

ON-SITE QUESTIONS

1. Data set `Smarket.csv` contains data on percentage returns for the S&P 500 stock index over 1250 days, from the beginning of 2001 until the end of 2005. For each date, the data contains the trading volume and the percentage returns for each of the five previous trading days, Lag1 through Lag5.

(c) Consider $\delta = 0.5184$ as the border line to predict the direction be up or down, calculate the accuracy of model M3 when predicting for the days in the data set given.

Hint: To recall on what the role of δ is, please revise Tutorial 4 question 2

Discuss with your group! Time limit: 3 mins

ON-SITE QUESTION 3

Q: Consider $\delta = 0.5184$ as the threshold to predict the direction be up or down, calculate the accuracy of model M3

Step 1: Get the Predicted Probability of each data point from M3

```
# Step 1: Probability of y = 1 (Direction = Up) for each day  
pred.M3 = predict(M3, newdata = market[,3:8], type = "response")
```

Don't forget type = "response"

```
> pred.M3[1:3]  
      1      2      3  
0.5070841 0.4814679 0.4811388
```

ON-SITE QUESTION 3

Q: Consider $\delta = 0.5184$ as the threshold to predict the direction be up or down, calculate the accuracy of model M3

Step 2: Classify data based on the threshold $\delta = 0.5184$

```
# Probability >= 0.5184 then that day is predicted as 'Up', else 'Down'  
pred.direction = ifelse(pred.M3 >= 0.5184, "Up", "Down")
```

Alternatively:

```
# Probability >= 0.5184 then that day is predicted as 1, else 0  
pred.direction = ifelse(pred.M3 >= 0.5184, 1, 0)
```

ON-SITE QUESTION 3

Q: Consider $\delta = 0.5184$ as the threshold to predict the direction be up or down, calculate the accuracy of model M3

Step 2: Classify data based on the threshold $\delta = 0.5184$

```
# Probability >= 0.5184 then that day is predicted as 'Up', else 'Down'  
pred.direction = ifelse(pred.M3 >= 0.5184, "Up", "Down")
```

```
> table(pred.direction)  
pred.direction  
Down      Up  
  631    619
```

ON-SITE QUESTION 3

Q: Consider $\delta = 0.5184$ as the threshold to predict the direction be up or down, calculate the accuracy of model M3

Step 3: Calculate Accuracy

```
accuracy = mean(pred.direction == market$Direction)
accuracy # 0.5352
```

Note: If you used 0, 1 in step 2, then you should compare with **market\$y** instead

ON-SITE QUESTION 3

Q: Consider $\delta = 0.5184$ as the threshold to predict the direction be up or down, calculate the accuracy of model M3

Step 3: Calculate Accuracy

```
# Alternatively: using confusion matrix to get accuracy
conf.mat = table(market$Direction, pred.direction)

accuracy = sum(diag(conf.mat))/sum(conf.mat)
accuracy # 0.5352
```

ON-SITE QUESTIONS

1. Data set `Smarket.csv` contains data on percentage returns for the S&P 500 stock index over 1250 days, from the beginning of 2001 until the end of 2005. For each date, the data contains the trading volume and the percentage returns for each of the five previous trading days, Lag1 through Lag5.

(d) (*Extra*) Write R code to plot the ROC curve for model M3. Derive and report AUC value of it.

Discuss with your group! Time limit: 2 mins

ON-SITE QUESTION 4

Q: Write R code to plot the ROC curve for model M3. Derive and report AUC value of it.

```
library(ROCR)

roc.obj = prediction(pred.M3, market$y) # rmb pred.M3 is alr prob of y=1!

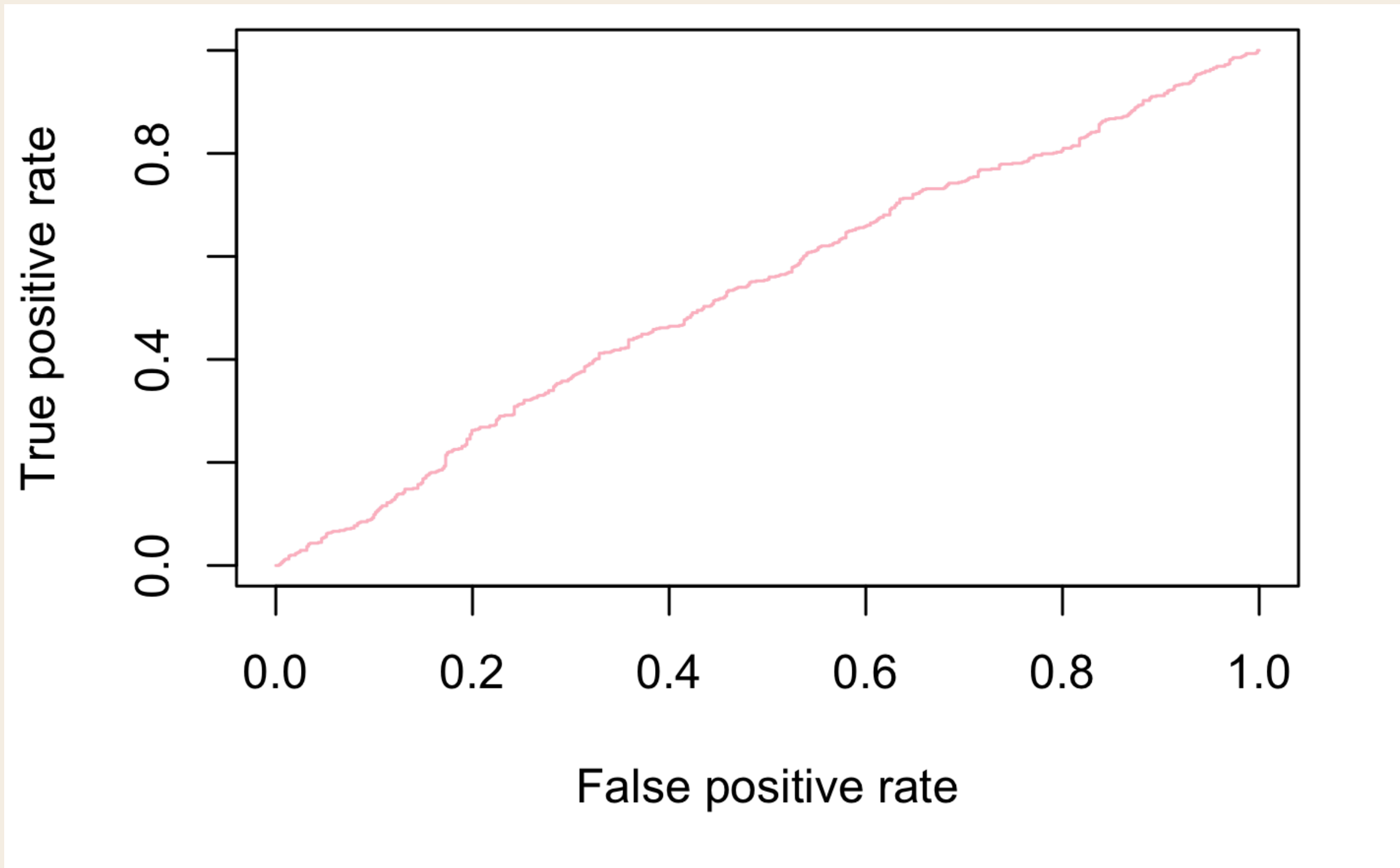
roc = performance(roc.obj, measure = 'tpr', x.measure = 'fpr')

plot(roc, col = 'pink')

auc = performance(roc.obj, "auc")@y.values[[1]]
auc # 0.5387 (not much more ideal than random...)
```

ON-SITE QUESTION 4

Q: Write R code to plot the ROC curve for model M3. Derive and report AUC value of it.



OFF-SITE QUESTIONS

Let's go to code!

ASSIGNMENT 2

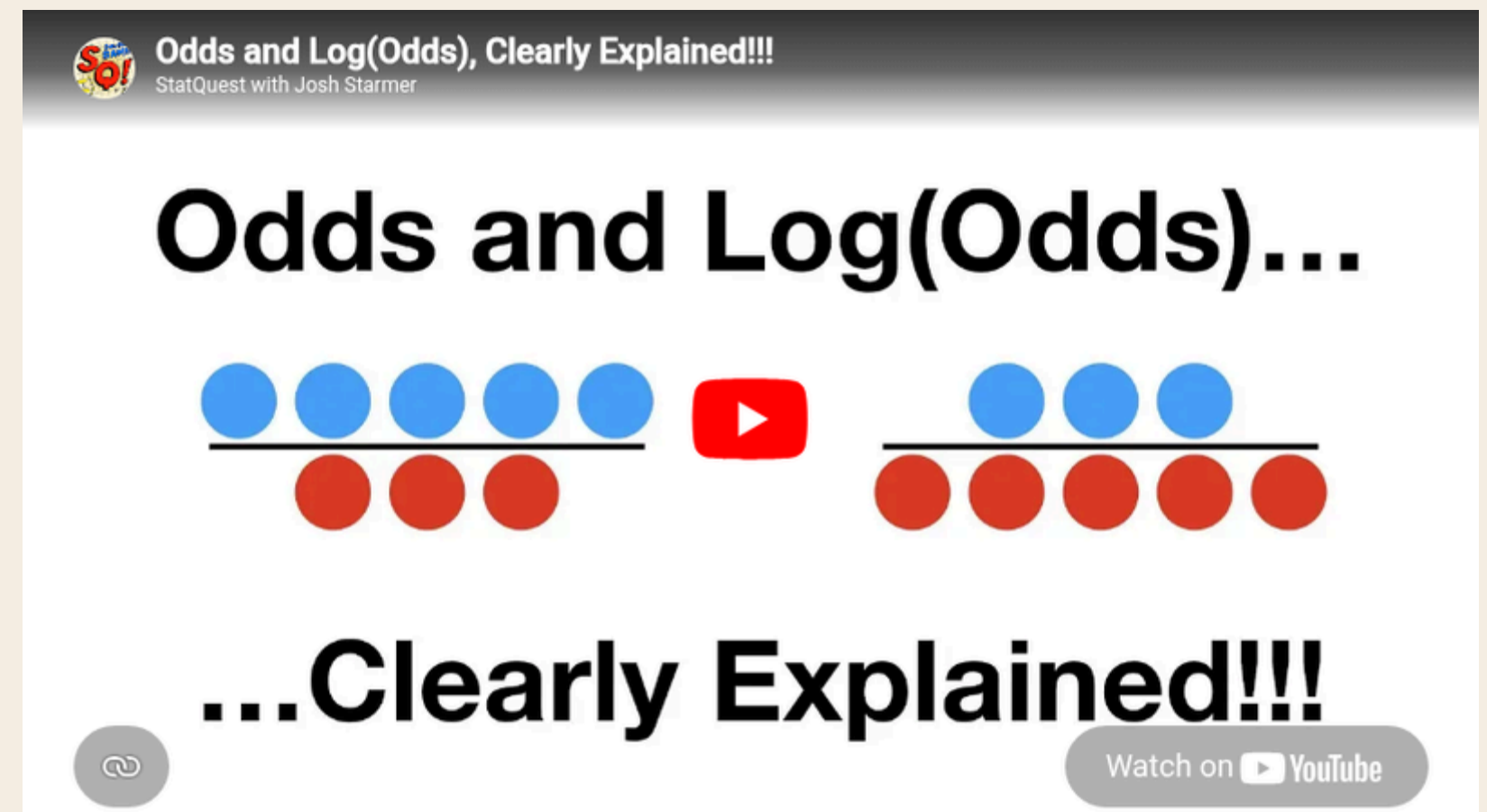
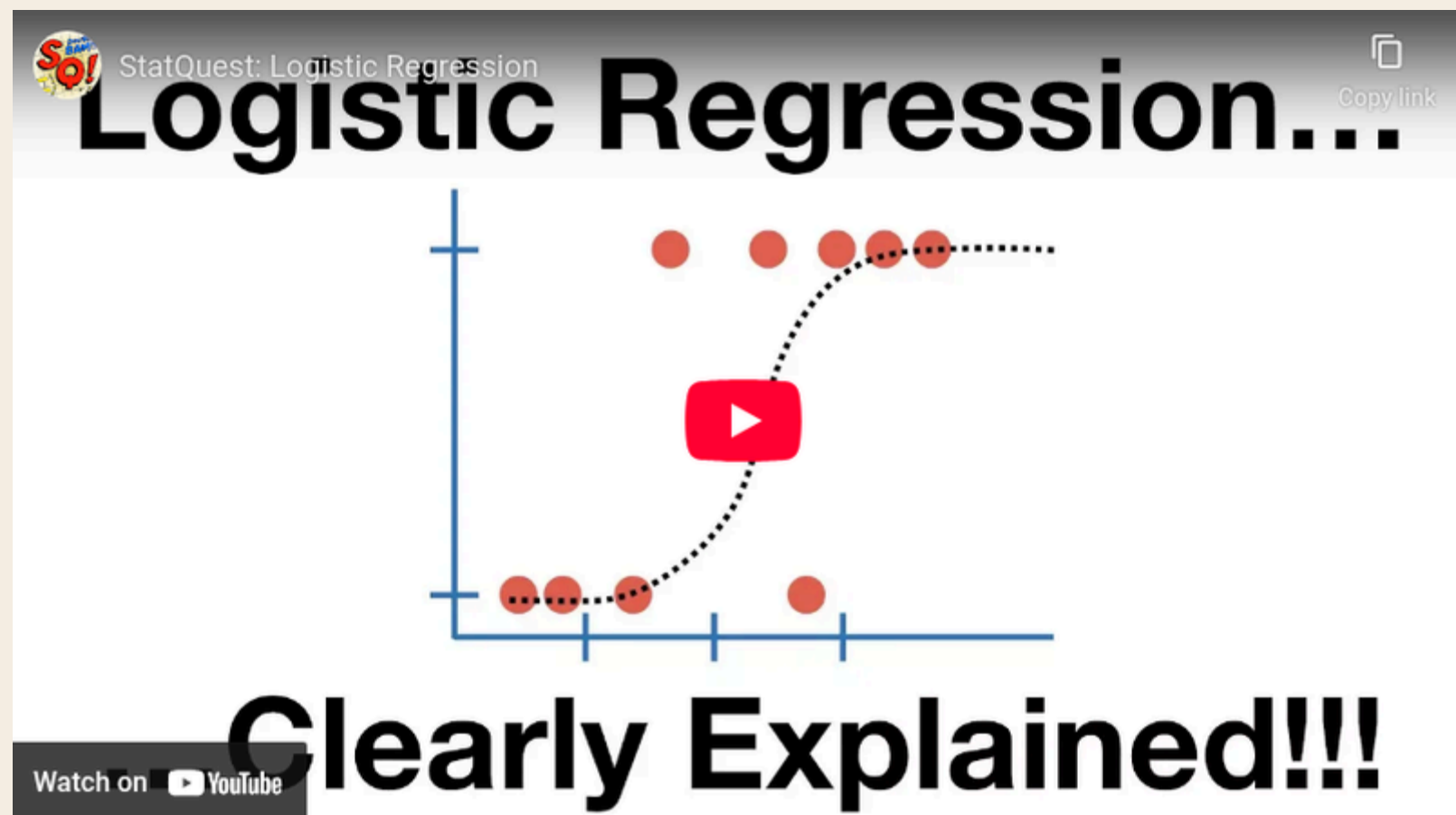
TIPS

Extra!!!

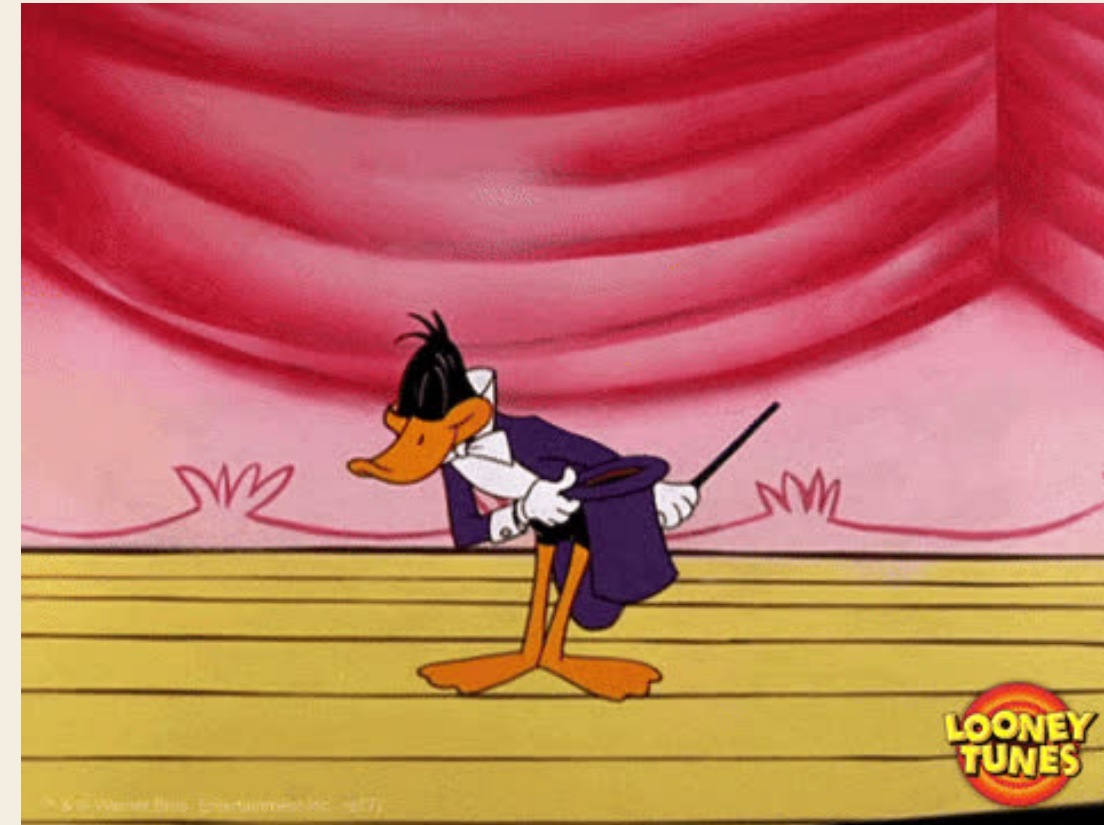
Videos of Naive Bayes (by StatQuest):

- *Logistic Regression*: <https://www.youtube.com/watch?v=yIYKR4sgzI8>
- *Odds and Log(Odds), Clearly Explained!!!*:

<https://www.youtube.com/watch?v=ARfXDSkQf1Y>



THANK YOU!



Weekly Slides & Code will be shared on Github (star it for
bookmark):

<https://github.com/kr7001/DSA1101-teaching-materials>